

Interface Capability Brief

EDI 837 NCCI PTP · Stakeholder overview · 2026-08-13

1. Executive summary

Show that PilotFish can apply CMS Procedure-to-Procedure (PTP) NCCI edits as a business-rule layer on X12 837P claims: look up two procedure codes on the same encounter against an external PTP table and kick out pairs that should not be billed together unless an allowed modifier is present. There is no stock PilotFish NCCI/PTP module. Med Rec implements MUE (units per CPT) for HL7 DFT, not PTP pairs — this demo is the pair half of Karthik's ask.

2. Who this is for

- Sources: Directory-drop 837P (input/inbound/*.edi) - Destinations: output/pass/ (no pair, or modifier allowed), output/kickout/ (NCCI pair without a valid exception) - Demo vs production: Demo only — two synthetic PTP rows (not official CMS quarterly files). Production would load CMS PTP CSVs into SQL on the CMS publish cycle. - Sibling demo: edi-837-ncci-mue (units vs max). Same 837 intake pattern.

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File (DirectoryListener)
- Format / envelope: X12 837P 005010X222A1, //Transaction fork
- Identity fields: CLM01; first two SV1 CPT codes; modifier on the second SV1 (SV101-03 / HC:cpt:59)
- Samples path: samples/sample_837p_ptp.edi

Integration highlights

FQCN sources: edi-837-ncci-mue, edi-835-payment-integrity, CMS PTP modifier indicator (0/1/9). Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	com.pilotfish.eip.modules.file.DirectoryListener	Poll inbound. Skeleton: edi-837-ncci-mue
R1 Format	EDITransformationModule + XPathForkingModule //Transaction	TableData edi-tabledata/837-Q1 (5010)
R1 Target processors	XPath identity → XSLT pair extract → DatabaseSqlProcessor PtpEdits → XSLT decision	Symmetric Column1/Column2 lookup
R1 Transport	DirectoryTransport + XML Formatting on the transport	Stage under output/staged-decisions/
R2 Listener	DirectoryListener	Poll staged decisions
R2 Router	XPathRoutingModule accumulate=true	//MatchBucket = pass / kickout

Stage	Module / mechanism	Notes
R2 Transports	DirectoryTransport	Unique Pretty-Print names per target

4. Honest boundaries (not claimed yet)

- Load a real CMS PTP quarterly extract in a later pass?
- Evaluate all SV1 combinations, not just the first two lines?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Poll And Score PTP” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: EDI parses; first two SV1 CPTs looked up in PtpEdits; modifier indicator vs modifier 59/X{E,P,S,U} - What is NOT checked: SNIP 1–7, MUE units (sibling demo), full CMS PTP file, later-line combinations beyond the first two SV1s, medically necessary exception documentation - Does failure block outbound? yes for unparseable EDI; kickout theater for PTP hits

6. State & idempotency

- Status model: none (stateless file buckets). Catalog is read-only seed. - When state advances: after PtpDecision is staged and routed - Dedup keys: CLM01 (overwrite same filename on resubmit) - Retry / poison: unparseable 837 fails at EDI→XML (fail-closed)

8. Dual-write / side effects

- Order: decision XML → route to pass or kickout file - Compensation: none - Demo shortcuts: two synthetic pairs; original 837 is not re-emitted

3. Inbound contract

- Transport: Directory / File (DirectoryListener) - Format / envelope: X12 837P 005010X222A1, //Transaction fork - Identity fields: CLM01; first two SV1 CPT codes; modifier on the second SV1 (SV101-03 / HC:cpt:59) - Samples path: samples/sample_837p_ptp.edi

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Poll And Score PTP: runs steps such as Poll EDI 837P Files, Fork - XPath, Stage PTP Decision.

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2 - Route PTP Decisions: runs steps such as Poll Staged PTP Decisions, Conditional Router, Write Pass, Write Kickout.

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7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — No stock NCCI module — Product has SNIP, not CMS PTP tables
- Risk noted in design: Med — 23R1 trial X12 tables expired — Sandbox mounts EDI/TableData/x12
- Risk noted in design: Med — Official CMS quarterly files not loaded — Demo seed only
- Risk noted in design: Med — Duplicate processor names — EIP refuses the route
- Risk noted in design: Low — Em-dash / non-ASCII in SQL Description — XML Formatting + XPath router reject 0x14

8. Seeing it work

- - Ports: SQL 14343, EIP 8132, Web UI 8133 - Volumes: ./input, ./output, ./logs, ./samples, ./documents, TableData mount - Heap: 512M–2GB - Dependencies / cold start: SQL health + seed ~30–60s, EIP ~60–90s - Credentials: sa / PilotFish_Demo1! · Edi837NcciPtp
- Local: http://127.0.0.1:8133/
- EIP: http://127.0.0.1:8132/eip/

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- SQL: 127.0.0.1:14343 (sa / PilotFish_Demo1! · Edi837NcciPtp)
 - docker compose --profile full up -d --build

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.